

torch.functional.cartesian_prod#

[https://pytorch.org/docs/stable/generated/torch.functional.cartesian_prod.ht](https://pytorch.org/docs/stable/generated/torch.functional.cartesian_prod.html)

Description:

Do cartesian product of the given sequence of tensors. The behavior is similar to python's `itertools.product`. *tensors (Tensor) – any number of 1 dimensional tensors. A tensor equivalent to converting all the input tensors into lists, do `itertools.product` on these lists, and finally convert the resulting list into tensor.

Function Signature

```
torch.functional.cartesian_prod(*tensors)[source]#
```

Parameters

Returns

Return type

Returns:

Tensor

Code Examples

```
>>> import itertools
>>> a = [1, 2, 3]
>>> b = [4, 5]
>>> list(itertools.product(a, b))
[(1, 4), (1, 5), (2, 4), (2, 5), (3, 4), (3, 5)]
>>> tensor_a = torch.tensor(a)
>>> tensor_b = torch.tensor(b)
>>> torch.cartesian_prod(tensor_a, tensor_b)
tensor([[1, 4],
        [1, 5],
        [2, 4],
        [2, 5],
        [3, 4],
        [3, 5]])
```

Detailed Documentation

Documentation

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